

भारतीय मानक
Indian Standard

IS 19791 : 2026

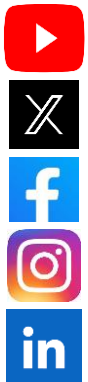
**ईएनटी सर्जरी उपकरण —
माइक्रोडेब्राइडर — विशिष्टि**

**ENT Surgery Instruments —
Microdebrider — Specification**

ICS 11.040.30

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भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS
मानक भवन, 9 बहादुर शाह ज़फर मार्ग, नई दिल्ली - 110002
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI - 110002
www.bis.gov.in www.standardsbis.in

April 2026

Price Group 5

Ear, Nose, Throat and Head & Neck Surgery (ENT - H&N) Instruments Sectional Committee, MHD 04

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards after the draft finalized by the Ear, Nose, Throat and Head & Neck Surgery (ENT - H&N) Instruments Sectional Committee had been approved by the Medical Equipment and Hospital Planning Division Council.

A microdebrider is a powered surgical instrument used mainly in ENT procedures, for removal of soft tissue. It employs a motor-driven, hollow rotating or oscillating blade combined with continuous suction, allowing tissue to be cut and aspirated simultaneously. Microdebriders are commonly used in FESS, nasal polypectomy, turbinoplasty, and adenoidectomy, and are available with various blade designs to suit different anatomical needs.

Microdebrider subject was taken up for standard formulation under the directions of Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India as part of implementation of National medical device policy, 2023.

The composition of the Committee responsible for formulation of this standard is given in [Annex B](#).

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be same as that of the specified value in this standard.

Indian Standard

ENT SURGERY INSTRUMENTS — MICRODEBRIDER — SPECIFICATION

1 SCOPE

This standard provides requirements of a Microdebrider system used for resecting tissues and soft bones. This standard also provides requirements for irrigation system that comes along with Microdebrider system.

2 REFERENCE

The standards listed in [Annex A](#) contain provisions, which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards.

3 TERMS AND DEFINITIONS

For the purpose of this standard, the following terms and definitions shall apply

3.1 Microdebrider — Electrically powered surgical instrument intended to cut, shave, or remove tissue using a rotating or oscillating cutting accessory with an integrated suction channel.

3.2 Microdebrider Hand Engine — Powered handheld surgical instrument intended to drive cutting accessories, such as blades and burs, for tissue or bone resection.

NOTE — The hand engine can operate in continuous rotation, forward/reverse rotation, or oscillation modes, as specified by the manufacturer.

3.3 Power Console — Electrical control unit intended to supply power to and control the operation of the microdebrider hand engine, including speed, direction, irrigation, suction, and associated safety functions.

3.4 Foot Control Unit — User-operated control device intended to enable hands-free control of the microdebrider operating parameters, such as speed and direction or mode of rotation.

3.5 Blades — Cutting accessories intended for resection of soft tissue and thin bone.

3.6 Burs — Rotary bone-cutting accessories intended for drilling, undercutting, or reshaping

bone.

3.7 Oscillation Mode — Operating mode in which the cutting accessory alternately rotates in clockwise and counter-clockwise directions.

3.8 Suction Interface — Connection point intended to remove resected tissue and fluids from the surgical site during operation.

3.9 Irrigation Pump — Component intended to deliver irrigation fluid to the surgical site to cool cutting accessories and reduce thermal effects on tissue or bone during operation.

4 REQUIREMENT

4.1 The Microdebrider system including the console, handpiece, footswitch and accessories, shall conform to the electrical safety requirements specified in IS/IEC 13450 (Part 1).

4.2 All components other than blades and burs cutting window shall free from sharp edges and other defects that could affect safety or usability. For guidance on design and usability considerations, the manufacturer may refer to IS 17922 (Part 1) and IS/IEC/TR 62366-2.

4.3 All patient-contacting materials shall be evaluated for biocompatibility in accordance with IS 17932 (Part 1).

4.4 Reusable components shall be capable of withstanding the intended cleaning, disinfection and sterilization cycles without degradation of their structural integrity or functional performance.

4.5 The foot-actuated control shall provide the required RPM output within a tolerance of ± 5 percent of the manufacturer-specified value and shall be tested in accordance with the manufacturer's guidelines.

4.6 The system should prevent operation when the cutter is exposed in an unsafe condition or when attachments are not correctly seated. The device should incorporate overload/over-torque protection to stop rotation and provide an appropriate user alert. For guidance on design and usability considerations, the manufacturer may refer to IS 17922 (Part 1) and IS/IEC/TR 62366-2.

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4.7 The handpiece should be lightweight, ergonomically balanced and easy to maneuver during prolonged surgical procedures. It shall be evaluated using appropriate phantom models to assess grip and potential clogging.

4.8 All controls shall be designed to minimize the risk of accidental activation and shall remain easy to operate while wearing surgical gloves.

4.9 The Microdebrider system should be compatible with endoscopes, vacuum pumps, sterile irrigant bags or syringes, and collection canisters, as well as any additional devices as specified in the manufacturer's instructions for use.

5 FUNCTIONAL REQUIREMENT

The essential components of the microdebrider system shall conform to the following requirements.

5.1 Microdebrider Hand Engine

5.1.1 Dimensional Requirements

- a) The diameter of hand engine shall be within the range of 14mm to 25mm, to enable user to conveniently hold and use to give adequate comfort and grip to the user; and
- b) The length of the hand engine shall not be more than 300 mm.

5.1.2 Suction Requirements

The suction system shall comply with IS 18375 (Part 1).

5.1.3 Attachment Locking System

- a) The blade and bur mounting interface shall include a durable and repeatable locking mechanism complying with manufacturer specifications; and
- b) Locking shall withstand operational torque and axial loads without unintended loosening.

5.2 Power Console

5.2.1 Reverse direction operation shall trigger visible and audible indication on the console.

5.2.2 The power console should incorporate with LCD touch display, supporting speed setting, operating mode selection, irrigation/suction control and operational status display.

5.2 Foot Control Unit

The foot control shall have ingress protection as

specified for remote foot switches with electrical switching parts in IS 18375 (Part 1).

5.3 Shaver Blades (Straight and Angled)

5.4.1 The shaver blade may be provided in one or more of the following configurations:

- a) Rounded tip;
- b) Side-facing cutting window with serrated edges;
- c) Fixed curved profile;
- d) Malleable (bendable) design; and
- e) Rotatable blade design.

5.4.2 All blades should have irrigation port/nozzle enabling directed flow for thermal management.

5.4.3 Each blade shall integrate suction flow paths compatible with the handpiece suction lumen.

5.5 Burs

5.5.1 Burs shall withstand maximum rated rotational speed without fracture when tested with appropriate phantom models.

5.5.2 All burs shall include an irrigation feature for cooling and debris removal (wherever applicable).

6 TEST

6.1 Basic Safety and Essential Performance

The complete system (console, handpiece, foot control, and attachments) shall be tested for electrical safety as per IS/IEC 13450 (Part 1).

6.2 The device shall be tested for electromagnetic compatibility according to IS 13450 (Part 1/Sec 2).

6.3 All components shall undergo biocompatibility testing based on the evaluation performed in accordance with [4.2](#), using the test methods prescribed in the applicable parts of the IS 17932 series of standards.

6.4 Corrosion Resistance Metallic

Stainless steel components including handpiece, blades and burs shall be tested for corrosion resistance as per IS 7531.

6.5 Drop and Mechanical Durability

Console, handpiece, cables and footswitch shall undergo mechanical durability and drop testing under simulated clinical handling conditions without loss of usability or electrical safety.

6.6 Clogging Resistance Test

A clogging resistance test shall be performed to validate the ability of the cutting blades and suction channels to handle typical sinus and soft-tissue loads. The test shall be conducted using tissue-simulating phantom material. The handpiece, fitted with the cutting blade, shall be connected and operated at the maximum and minimum rotational speeds for 15 min, in both forward and reverse directions. Conformity shall be assessed by visual inspection to verify the presence or absence of clogging of the blade and/or suction channels during or after the test. The blade and/or suction channels shall not become clogged.

The system shall be tested to ensure that the suction flow rate remains within ± 5 percent of the selected flow rate when operated at the maximum and minimum rated flow rates.

6.7 Rotational/Oscillating Speed Accuracy Test

The system shall be tested to ensure that forward rotation, reverse rotation and oscillating modes deliver speeds within ± 5 percent of the selected speed, when operated at the maximum and minimum rated speeds with speed stability maintained even under simulated soft-tissue cutting loads.

6.8 Irrigation Flow-Rate and Distribution Test

For blades and burs equipped with irrigation ports, testing shall confirm that coolant is adequately delivered to the cutting tip, flow distribution is uniform, and that irrigation effectively supports debris evacuation, using clinically representative irrigation volumes. The delivery flow rate should be ± 5 percent of the selected flow rate when operated at the maximum and minimum rated flow rates.

6.9 Handpiece Vibration and Noise-Level Test

Handpiece performance shall be assessed by measuring vibration amplitude during maximum speed operation (as declared by manufacturer) and verifying that noise levels remain within the manufacturer's specified limits, not exceed 75 dB(A) at a distance of 1.0 m, to ensure surgeon

comfort and safe tissue handling (Manufacturers may refer to IS 18375 (Part 4) for more information on noise test).

7 MARKING

7.1 Each Microdebrider system must be labeled with the following information:

- a) Manufacturer's name/trademark;
- b) Name and address of the manufacturer;
- c) Declared minimum and maximum speed (in rpm);
- d) Declared minimum and maximum flow rates;
- e) Declared maximum noise limited (in dBA);
- f) Name and address of the marketer (if applicable);
- g) Month and year of manufacture; and
- h) Unique Device Identification (UDI)/serial number.

7.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

8 PACKING

The device and its accessories must be securely packed within a case or box to ensure that they remain intact and protected during storage and transportation.

9 INFORMATION TO BE SUPPLIED BY THE MANUFACTURER

The manufacturer should provide information for identification and labelling, including packaging, and accompanying information, in accordance with IS/ISO 20417.

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ANNEX A

(Clause 2)

LIST OF REFERRED STANDARDS

IS No.	Title	IS No.	Title
IS 7531 : 1990	Surgical instruments — Corrosion resistance of stainless steel surgical instruments — methods of tests (<i>first revision</i>)		Section 2 Electromagnetic disturbances — Requirements and tests (IEC 60601-1-2 : 2020, MOD (<i>second revision</i>))
IS/ISO 14971 : 2019	Medical devices — Application of risk management to medical devices (<i>first revision</i>)	IS 17932 (Part 1) : 2023	Biological evaluation of medical devices: Part 1 Evaluation and testing within a risk management process (ISO 10993-1 : 2018, MOD)
IS 13450 (Part 1) : 2024	Medical electrical equipment: Part 1 General requirements for basic safety and essential performance (IEC 60601-1 : 2020, MOD) (<i>third revision</i>)	IS 18375 (Part 1) : 2023	Medical suction equipment: Electrically powered suction equipment (ISO 10079-1 : 2022, MOD)
IS 13450 (Part 1/ Sec 2) : 2024	Medical electrical equipment: Part 1 General requirements for basic safety and essential performance,	(Part 4) : 2025	General requirements (ISO 10079-4 : 2021, MOD)
		IS/IEC 60529 : 2001	Degrees of protection provided by enclosures (IP code)

To access Indian Standards click on the link below:
<https://standards.bis.gov.in/website/know-your-standards>

ANNEX B

(Foreword)

COMMITTEE COMPOSITION

Ear, Nose, Throat and Head & Neck Surgery (ENT - H&N) Instruments Sectional Committee, MHD 04

<i>Organization</i>	<i>Representative(s)</i>
All India Institute of Medical Sciences, New Delhi	DR RAKESH KUMAR (Chairperson)
All India Institute of Medical Sciences, New Delhi	DR PREM SAGAR DR ARVIND KUMAR (<i>Alternate I</i>) DR KAPIL SIKKA(<i>Alternate II</i>)
ALPS International, New Delhi	SHRI ALOK NARANG SHRI KARAN NARANG (<i>Alternate</i>)
All India Institute of Medical Sciences, Bhopal	DR VIKAS GUPTA DR GANAKALYAN BEHERA (<i>Alternate</i>)
Association of Indian Medical Device Industry, New Delhi	SHRI TARLOCHAN DEV SHRI ANKUR BHARGAVA (<i>Alternate I</i>) DR C. S. PRASAD (<i>Alternate II</i>)
Directorate General of Health Services, New Delhi	DR PALLIKA KUMAR
Government Medical College & Hospital, Chandigarh	DR SURINDER K. SINGHAL DR NITIN GUPTA (<i>Alternate</i>)
Happy Reliable Surgeries Private Limited, Bengaluru	SHRI HEMANT SAVALE SHRI SANJEEV GAUTAM (<i>Alternate</i>)
Indian Institute of Technology Kanpur, Kanpur	DR A. R. HARISH
Indian Institute of Technology Guwahati, Guwahati	DR S. KANAGARAJ
Kalam Institute of Health Technology, Vishakhapatnam	DR ARJUN THIMMAIAH SHRI AMIT SHARMA (<i>Alternate</i>)
Maulana Azad Medical college, New Delhi	DR RAVI MEHER
Serwell MediEquip, Chennai	SHRI T. JEBIN SAMUEL SHRI R. RADHAKRISHNAN (<i>Alternate I</i>) SHRI G. SATHISH (<i>Alternate II</i>)
Tata Memorial Center (Hospital), Mumbai	DR RICHA VAISH SHRI VIJAY YASHWANT MESTRI (<i>Alternate I</i>) DR ARJUN SINGH (<i>Alternate II</i>)
Post Graduate Institute of Medical Education and Research, Chandigarh	DR JAIMANTI BAKSHI
Eon Meditech Pvt Ltd, Surat	SHRI NIRAV PRAVINBHAI KHUNT SHRI MANISH DONASIYA (<i>Alternate</i>)
Stryker India Pvt Ltd, Gurugram	DR PRACHI PAL SHRI GAUTHAM (<i>Alternate</i>)
BIS Directorate General	SHRI ADITYA DAS SCIENTIST 'E'/DIRECTOR AND HEAD (MEDICAL EQUIPMENT AND HOSPITAL PLANNING) [REPRESENTING DIRECTOR GENERAL (<i>Ex-officio</i>)]

Member Secretary

SHRI KARTHIK REDDY KATIPALLY
SCIENTIST 'B'/ASSISTANT DIRECTOR
(MEDICAL EQUIPMENT AND HOSPITAL PLANNING), BIS

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Rhinology (Nose), MHD 04/WP 04

<i>Organization</i>	<i>Representative(s)</i>
All India Institute of Medical Sciences, New Delhi	DR RAKESH KUMAR (Convener)
All India Institute of Medical Sciences, New Delhi	DR PREM SAGAR DR ARVIND KUMAR (<i>Alternate</i>)
Government Medical College & Hospital, Chandigarh	SHRI SURINDER K. SINGHAL
India Medtronic Private Limited, Gurugram	SHRI SAURABH SABLE SHRI GURPREET SINGH(<i>Alternate I</i>) SHRI SANDEEP VERMA(<i>Alternate II</i>)
Karl Storz Endoscopy India Private Limited, New Delhi	SHRI SANDEEP SETHI SHRI KAPIL RANA (<i>Alternate</i>)
Serwell MediEquip, Chennai	SHRI T. JEBIN SAMUEL
Stryker India Private Limited, Gurugram	DR PRACHI PAL SHRI GAUTHAM A. (<i>Alternate</i>)

Microdebrider Working Group MHD 04/WP 04/WG 01

<i>Organization</i>	<i>Representative(s)</i>
Serwell MediEquip, Chennai	SHRI T. JEBIN SAMUEL (Convener)
Government Medical College & Hospital, Chandigarh	DR SURINDER K. SINGHAL
Stryker India Private Limited, Gurugram	DR PRACHI PAL

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Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002

Telephones: 2323 0131, 2323 3375, 2323 9402

Website: www.bis.gov.in

Regional Offices:

	Telephones
Central : 601/A, Konnectus Tower -1, 6 th Floor, DMRC Building, Bhavbhuti Marg, New Delhi 110002	{ 2323 7617
Eastern : 8 th Floor, Plot No 7/7 & 7/8, CP Block, Sector V, Salt Lake, Kolkata, West Bengal 700091	{ 2367 0012 2320 9474
Northern : Plot No. 4-A, Sector 27-B, Madhya Marg, Chandigarh 160019	{ 265 9930
Southern : C.I.T. Campus, IV Cross Road, Taramani, Chennai 600113	{ 2254 1442 2254 1216
Western : 5 th Floor/MTNL CETTM, Technology Street, Hiranandani Gardens, Powai Mumbai 400076	{ 25700030 25702715

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